



University of  
Central Lancashire  
UCLan Cyprus



# **ΣtatMod2026**

## **Statistical Modelling with Applications**

**Conference Programme**

18-19 September 2026  
UCLan Cyprus, Larnaka, Cyprus

## Welcome Message

On behalf of the Organising Committee, it is our great pleasure to welcome you to *StatMod2026: Statistical Modelling with Applications*, held at UCLan Cyprus in Larnaka on 18-19 September 2026.

Statistical modelling continues to play a fundamental role in addressing scientific and societal challenges across a broad spectrum of disciplines, including data science, artificial intelligence, engineering, finance, health sciences, environmental sciences and reliability. The aim of StatMod2026 is to provide an international forum where researchers and practitioners can present recent advances, exchange ideas and establish new collaborations.

This year's conference brings together participants from universities, research centres and industry across Europe and beyond. The programme reflects the breadth of contemporary statistical research, ranging from theoretical developments to methodological innovations and real-world applications.

The Organising Committee would like to express its sincere appreciation to all invited speakers, contributed speakers, members of the Steering Committee, Scientific Committee and reviewers for their invaluable support. We also thank our sponsors for their support and UCLan Cyprus for hosting the conference and providing an excellent environment for scientific exchange.

We hope that you will enjoy both the scientific programme and the opportunity to meet colleagues from around the world. We wish you a productive conference and a pleasant stay in Cyprus.

The Organising Committee

## Sponsors

The Organising Committee gratefully acknowledges the generous support of our sponsors. Their contribution has played an important role in the successful organisation of the **StatMod2026 Conference** and is sincerely appreciated.

We thank all our sponsors for their commitment to supporting research, education, and collaboration within the mathematical and statistical sciences.

### Gold Sponsors



### Silver Sponsors



| Friday, September 18th |                                                                                                                                                                                                               |                                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 9:00 - 10:00<br>Lobby  | Conference registration                                                                                                                                                                                       |                                                                                                                                           |
| 10:00 - 10:10<br>AUD2  | Opening Session                                                                                                                                                                                               |                                                                                                                                           |
|                        | <b>Vlad S. Barbu</b> , Co-chair of the $\Sigma$ statMod Conference Series<br>Welcome Address<br><br><b>Milto Hadjikyriakou</b> , Chair of the Organising Committee of $\Sigma$ statMod2026<br>Welcome Address |                                                                                                                                           |
| 10:10- 11:10<br>AUD2   | Plenary Session                                                                                                                                                                                               |                                                                                                                                           |
|                        | Chair: Vlad S. Barbu                                                                                                                                                                                          |                                                                                                                                           |
|                        | <b>Davy Paindaveine</b> , Stopping on the last success with unknown odds                                                                                                                                      |                                                                                                                                           |
| 11:10-11:40<br>Lobby   | Coffee break - Group Photo                                                                                                                                                                                    |                                                                                                                                           |
| 11:40 -13:10           | <b>Statistical Methodology and Inference</b><br>Chair: Bojana Milošević<br>Room CY006                                                                                                                         | <b>Information Measures and Statistical Divergences</b><br>Chair: Vlad S. Barbu<br>Room CY013                                             |
|                        | <b>Argyro Damoulaki</b> , Bounding posterior evidence for variable screening                                                                                                                                  | <b>Aida Toma</b> , Robust tests for moment condition models                                                                               |
|                        | <b>Dounia BOURZIK</b> , Detecting bilinear dependence in financial panel data: a rank-based approach                                                                                                          | <b>Razvan-Cornel Sfetcu</b> , Tsallis and Kaniadakis divergences between probability measures                                             |
|                        | <b>Marija Cuparić</b> , Testing independence under a monotone missingness surrogate model                                                                                                                     | <b>Elena-Corina Cipu</b> , Measuring population complexity through entropy measures                                                       |
|                        | <b>Ionel Popescu</b> , Convergence of PSO and the non-Gaussian limit                                                                                                                                          | <b>Silvia Dedu</b> , Risk models in actuarial science using entropy measures                                                              |
|                        | <b>Angeliki Boutsis</b> , Exponentiality tail test: a U-statistics approach                                                                                                                                   | <b>Mihaela Irma Vladescu</b> , Transition to a green economy in Romania - population perspectives: perceptions, barriers and expectations |
|                        | <b>Bojana Milošević</b> , A characteristic function-based test for the multi-sample problem under random right censoring                                                                                      | <b>Vlad S. Barbu</b> , On the weighted $(\Phi, a)$ power divergence family                                                                |

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|--------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13:10 -14:25 | <b>Lunch</b>                                                                                                                |                                                                                                                                                                                         |
| 14:25 -15:40 | <b>Stochastic Models for Risk, Finance and Decision-Making</b><br><b>Chair: Salvatore Vergine</b><br><b>Room CY006</b>      | <b>Statistics for Complex Data</b><br><b>Chair: Sophia Loizidou</b><br><b>Room CY013</b>                                                                                                |
|              | <b>Filippo Petroni</b> , Stochastic reliability indicators for wind energy systems: a Markov and semi-Markov perspective    | <b>Katarzyna Szczerbea</b> , How to analyse survival data with AI-informed non-linear Cox model                                                                                         |
|              | <b>Lucianna Cananà</b> , Optimal stopping valuation of modular advertising strategies under copula-based Markov dependence  | <b>Andreas Anastasiou</b> , Tensor time series change-point detection in cryptocurrency network data                                                                                    |
|              | <b>Guglielmo D'Amico</b> , Information theory and spatial risk analysis                                                     | <b>Chiara Passamonti</b> , A nonparametric approach for noisy compositional data                                                                                                        |
|              | <b>Michele Palma</b> , A factor-driven shared hidden Markov model for high frequency assets returns                         | <b>Christophe Ley</b> , An overview of Principal Component Analysis methods for directional data                                                                                        |
|              | <b>Salvatore Vergine</b> , Indexed Markov chains for intraday trading with technical indicators                             | <b>Sophia Loizidou</b> , Multiple change-point detection on the circle via isolation using permutation testing                                                                          |
| 15:40 -16:10 | <b>Coffee break</b>                                                                                                         |                                                                                                                                                                                         |
| 16:10 -17:25 | <b>Probability Theory, Stein's Method and Random Structures</b><br><b>Chair: Demetres Christofides</b><br><b>Room CY006</b> | <b>Applied Statistical Modelling</b><br><b>Chair: Andreas Makrides</b><br><b>Room CY013</b>                                                                                             |
|              | <b>Robert Gaunt</b> , Stein's method for the matrix normal distribution                                                     | <b>Hassan OUKHOUYA</b> : Classification of climate change adaptation plans (CCAP/PACC) for new Brunswick municipalities using machine learning, deep learning, and generative AI models |
|              | <b>Ferdinand Rapin</b> , Sharp Stein factors for integrated Pearson distributions through kernel representation             | <b>Pavel Averin</b> : The statistical engine of constraint-based causal discovery: conditional independence tests in practice                                                           |
|              | <b>Manuela-Simona Cojoclea</b> , When is an observation truly an outlier? a probability-coordinate perspective              | <b>Adela Ionescu</b> , The influence of parameters on multivariate optimization in inverse modelling problems                                                                           |
|              | <b>Christina Savvidou</b> , Diameter thresholds of random Cayley Graphs - I                                                 | <b>Steliana Rodino</b> , Data-driven modelling of complex agro-environmental systems                                                                                                    |
|              | <b>Demetres Christofides</b> , Diameter Thresholds of Random Cayley Graphs - II                                             |                                                                                                                                                                                         |

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| 17:25 - 18:35<br>AUD2 | <b>Plenary Session</b>                                                                               |
|                       | Chair: Milto Hadjikyriakou                                                                           |
|                       | <b>Maria Kateri</b> , A Divergence-Based Framework for Statistical Modeling and Reliability Analysis |
| 19:30                 | <b>Dinner</b>                                                                                        |

| Saturday, September 19th |                                                                                                                                                               |                                                                                                                           |                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 10:00 -11:30             | <b>Markov Chains, Time Series &amp; Forecasting</b><br><b>Chair: Andreas Anastasiou</b><br><b>Room CY006</b>                                                  | <b>Advances in Stochastic Models and Methods with Applications</b><br><b>Chair: Andreas Makrides</b><br><b>Room CY013</b> | <b>Reliability, Acceptance Sampling and Risk Modelling</b><br><b>Chair: Milto Hadjikyriakou</b><br><b>Room CY018</b>      |
|                          | <b>Vasilios Kalaitzis</b> , Extending Grey-Markov forecasting through second-order transition dynamics for energy mix prediction                              | <b>Sandra Ferreira</b> , A variance-stabilizing framework for higher-order cumulant estimation                            | <b>Spiros Dafnis</b> , A new family of generalized r-modified reliability systems                                         |
|                          | <b>Dimitris Karlis</b> , Some spatio-temporal models for count data                                                                                           | <b>D.-Jacob Economides</b> , Perpetual american options in a jump-diffusion model under Poisson observation times         | <b>Christos Papadopoulos</b> , A refined approach to approximating the mean number of system repairs                      |
|                          | <b>Anna Nalpantidi</b> , On joint modelling of discrete and continuous time series                                                                            | <b>Dário Ferreira</b> , Modelling structural changes in random-effect variability for short longitudinal panels           | <b>Mohd Azri Pawan Teh</b> , Modelling new two-sided group chain acceptance sampling plans for Gamma distribution         |
|                          | <b>Leonidas Kordalis</b> , Fast matrix-convolution methods for semi-Markov modelling                                                                          | <b>Alex Karagrigoriou</b> , Evaluation of seismic activity via Poisson hidden Markov models                               | <b>Mohd Azri Pawan Teh</b> , Modelling new group chain acceptance sampling plans via the minimum angle method             |
|                          | <b>Adyda Ibrahim</b> , Evaluating multi-regime operational sensitivity in urban transit: A transition from discrete to Bayesian continuous-time Markov models | <b>Nikolaos Galanopoulos</b> , Extensions of Grey forecasting models for insurance loss prediction                        | <b>Milto Hadjikyriakou</b> , Stochastic ordering of dependent systems under transformation models and Archimedean copulas |
|                          | <b>Konstantinos Fokianos</b> , Inference and applications of multivariate integer-valued time series                                                          | <b>Andreas Makrides</b> , Discrete-time semi-Markov modelling of residual dynamics in Grey GM(1,1) forecasting            |                                                                                                                           |
| 11:30 -12:00             | Coffee break                                                                                                                                                  |                                                                                                                           |                                                                                                                           |
| 12:00- 13:00<br>AUD2     | Plenary Session                                                                                                                                               |                                                                                                                           |                                                                                                                           |
|                          | Chair: Konstantinos Fokianos                                                                                                                                  |                                                                                                                           |                                                                                                                           |
|                          | <b>Antonio Di Crescenzo</b> , Some results on random distortions and hazard-operator families                                                                 |                                                                                                                           |                                                                                                                           |
| 13:00 -14:15             | Lunch                                                                                                                                                         |                                                                                                                           |                                                                                                                           |

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| 14:15 - 15:30 | <b>Sequential Estimation Methods and Change-Point Detection for Statistical Models</b><br><b>Chair: Ouerdia Arkoun</b><br><b>Room CY006</b> | <b>Statistical topics for Markov and Semi-Markov processes</b><br><b>Chair: Vlad S. Barbu</b><br><b>Room CY013</b>                    |
|               | <b>Salim Bouzebda</b> , Asymptotic properties of semiparametric M-estimators with multiple change points                                    | <b>Ana-Maria Staicu</b> , Repeatedly spatially indexed multivariate functional data: an application to prostate cancer identification |
|               | <b>Alain Celisse</b> , Early stopping for avoiding wasting time with random features                                                        | <b>Maroua SOUIAH</b> , Maximum-likelihood estimation for censored semi-Markov chains                                                  |
|               | <b>Laura Dumitrescu</b> , Penalized sequential estimation for recurrent events with high-dimensional covariates                             | <b>Ayhar Chafiâa</b> , Kernel smoothing techniques for the transition function of semi-Markov systems                                 |
|               | <b>Jean-Yves Brua</b> , Sequential model selection for nonparametric autoregressive processes                                               | <b>Emmanouil Nektarios Kalligeris</b> , Mixture kernels for non-stationary time series                                                |
|               | <b>Ouerdia Arkoun</b> , Robust adaptive efficient estimation for autoregressive Big Data models                                             | <b>Aguêmon Léon TEHUNGUE</b> , Parameter estimation and state decoding in hidden drifting Markov models                               |
| 15:30 -15:45  | <b>Coffee break</b>                                                                                                                         |                                                                                                                                       |
| 15:45- 16:45  | <b>Plenary Session</b>                                                                                                                      |                                                                                                                                       |
|               | Chair: Andreas Artemiou                                                                                                                     |                                                                                                                                       |
|               | <b>Ana Colubi</b> , Assessing topic changes using Latent Dirichlet Allocation                                                               |                                                                                                                                       |
| 16:45         | <b>Closing Session</b>                                                                                                                      |                                                                                                                                       |
|               | Chair: Milto Hadjikyriakou                                                                                                                  |                                                                                                                                       |